

CM 1230 – Foundation of Artificial Intelligence

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BSc Hons in AI

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Preamble

- We are in an era of Intelligent Machines
- The subject that builds the intelligent machines is called Artificial Intelligence
- AI builds natural intelligence into machines
 - Examples: Learning, thinking, walking, language understanding, recognition of images,
- AI has changed the way people do things (shopping, communication, travel, get medicine, learning, meetings, security, childcare, ...) in real life
- AI can attend to many tasks beyond the biological limitations of humans.
 - handling huge amounts of parameters/data, dynamic, noisy, very low access and reaction time, emotions, reaching the unsafe location, ...)
- As a technology AI builds power and control together
- This makes AI very different from other subjects
- You have to compete with graduates who have done AI modules in Arts, Commerce, Technology, CS, CE, SE, IT, DS, Cyber security

Outline

1. Evolution of AI education
2. What is Artificial Intelligence?
3. Diverse views on AI
4. Influential areas for the development of AI
5. Developments in AI
6. Major AI Technologies
7. Concern for building AI solutions

1. Evolution of AI education (module degree)

- In the 1950s, AI modules appeared in BSc in Computer Sciences degrees
- Edinburg University in the UK established the very first AI Department in the world
- After 2010, when more AI modules were identified, AI specializations in Computer Science degrees (BSc in Computer science specializing in Artificial Intelligence) have been introduced. – two years of AI
- After 2015, BSc Hons in Artificial Intelligence emerged –four years of AI
- In 2018, Carnegie-Mellon University (CMU) established the first BSc Hons in AI degree, and the first Faculty of Artificial Intelligence in the world.
- First BSc Hons in AI degree in Sri Lanka was started in 2021 by

Why we need a Four-year AI degree

- Each subject area promotes a certain thinking (mindset)
 - Mathematics – measurements, calculation, patterns, logical thinking
 - Science (know why)– experimental based decision/explanations
 - Technology (know-how) - knowledge even without a science, oldest knowledge $F/A = P?$
 - Engineering – constructing artifacts/machines (material, profit, appearance, weight, length, height, shapes, color)
- The same science ($F=ma$) continue for many centuries making technological advancements
- Formal knowledge - subjects we learn with theories, models, etc.
- Informal knowledge (beliefs, experience, commonsense) for problem-solving(cooking, adding salt, selecting fruits, artworks,),

Why we need a Four-year AI degree

- AI mindset involves dealing with both formal and informal knowledge
 - a doctor uses formal knowledge from MBBS degree, and experience gained by practice
- So, AI graduates are also expected to have a different mindset
- Graduate with such knowledge cannot be produced by two years of AI knowledge in a degree like BSc in Computer Science/Engineering
- Therefore, four-year AI degrees are introduced to build the correct mindset in AI with the graduate

AI as a distinct area

- AI shows scientific and engineering features

- Therefore, what is AI?

- AI is

• Science	wave	Na	Nimal
• Engineering	particle	Cl	Jagath
• Both	both	Both	Both
• Non of the above (<u>Distinct area</u>)	non of them	Different (NaCl)	Non of them

- With two observations, four options are available (Four-fold logic)

Electron has shown - particle/wave

- When a person dies who goes to the next life – you/another person

- (you, another person, both, non of them)

- Combine Mathematics (Pure, Applied, Both, Non of them)

- In order to ensure a different mindset of AI from the first year, you will learn some modules: Fundamentals of Machines, Introduction to Intelligent Machines, Microprocessor-based System, Mathematics, and Statistics other than AI modules

Exercise 2

- When the number of AI modules increases with various broad areas like machine learning, natural language processing, Agent technology, etc. what would be the future of BSc Hons in AI degrees?

Computing Landscape in Sri Lanka

- In 1985, Colombo University (Prof. Samaranayake) started BSc in Computer Science
- In 1986, Moratuwa University (Prof. Induruwa) BSc in Computer Science and Engineering
- After that, many universities started Computer Science specialization in BSc degrees
- In 1998, Sri Lanka Association for Artificial Intelligence was established (SLAAI)
- In 2001, Peradeniya University, BSc in Computer Engineering
- In 2001, the University of Moratuwa established the Faculty of IT
- In 2002, the University of Colombo School of Computing (UCSC) established
- In 2008, the University of Moratuwa launched the MSc in AI degree
- In 2020, the University of Moratuwa launched BSc Hons AI
- SLASSCOM apex body of the computing industry in Sri Lanka

Short notes
Memory
Thinking
writing speed

2
30/04/2024

2. What is Artificial Intelligence?

- Influence on Gestation of AI
 - World War II
 - Artificial Neural Networks
 - Digital Computer (Transistor, Von-Newman architecture)
 - Turing's idea of computability
 - Turing's paper
 - Minsky and McCarthy
 - ...
- Distinct areas for Intelligent Machines
 - McCarthy's proposal
 - Building natural intelligence to machines
 - Name of Artificial Intelligence
- Why AI is so different?
 - AI can solve any kind of problem in general
 - Power and control are built into the technology itself (Autonomy/singularity)
 - Which intelligent feature made AI so interesting in the recent past?

Gestation for AI

- Turing Machine (1936) -Computability
- During World War II, many people suffered, so interested in developments
- Ending World War II (1945) – Note the contributions of Einstein and Turing
 - Einstein – nuclear energy (atomic bomb), Turing – Bombe cracked the Enigma code
- McCulloch and Pitts (1943) - Artificial Neural Networks a model of the brain
- Invention of the Transistor (Semiconductor) and Digital Computer (1948)
- Turing's Seminal Paper, Computing Machinery and Intelligence (1950)
- Minsky and McCarthy completed their PhD at Princeton in the 1950s.
- Involvement of Princeton (Einstein, Turing, Minsky, McCarthy), Carnegie Mellon (McCarthy), MIT (McCarthy), Stanford (McCarthy), IBM (Watson)

Distinct Area for Intelligent Machines

- Dartmouth Conference (1956) was held with participants from CMU, MIT, Stanford, and IBM
- John McCarthy (CMU) proposed a distinct subject area for Intelligent Machines
- McCarthy named the new area for Intelligent Machines as Artificial Intelligence
- John McCarthy defined AI as Science and Engineering to build Intelligent Machines
- AI strives to understand natural intelligence and to construct intelligent machines
- AI builds models of natural intelligence into machines
 - Natural Intelligence – Learning, Decision Making, pattern recognition, image reconstruction, dancing, design,
- AI does not talk of machine-generated artificial kind of intelligence

Exercise

- Some people did not like the term Artificial intelligence because it gives the notion of machine-created intelligence.
 - But what AI does is model/program natural intelligence into machines
- Does this mean AI cannot produce something that the programmer did not know? (can a machine be creative?)
 - Assume a civil engineering knowledge of bridges was programmed. If the program is asked to generate all designs of a bridge by giving the width, length, weather condition, and soil condition, on the place to construct the bridge. Of course, it can – many architectural programs generate new designs that are not seen by architects beforehand.
- Note: regarding accuracy, efficiency, and creativity computers are ahead of us.
- If not carefully handled, AI can supersede mankind.

Exercise

- Can a machine think?
- What is thinking? – tell me activities where thinking is required/involved?
 - Problem-solving (GPS)
 - Decision making
 - Playing a game (Deep Blue)
 - Mathematical calculations
 - Theorem proving

Homework

- Do a search on chronology of Artificial Intelligence
 - Compile a video to demonstrate the history of AI

Why AI is so different?

- AI has changed the way people work (machines/home/office), communicate (distributed/with humans/with machines), live (shopping, relaxation, health, transport, banking, entertainment, learning), fight (ships, drones, aircraft,)
- AI can model both
 - formal knowledge (theory, algorithms, rules, procedures)
 - informal knowledge (experience, beliefs, commonsense, non-algorithmic/data)
- AI can solve any kind of problem because humans can solve any problem
- AI can solve a problem in many ways because humans can do so
- AI can give more than one answer for a problem (paracetamol or aspirins)
- AI solutions can be built on existing systems (ERP, MIS, Fan, Camara)
- AI can generate evolvable solutions (improving efficiency, and accuracy over the time)
- AI has built Power and Control together (AI gives autonomous solutions)
- AI is better in tasks where people have biological limitations (a large amount of data, accuracy, efficiency, reaction time, access time,)

Main difference of AI

- Any technology has Power and Control, where power is with the technology (Electricity, Car, nuclear energy, solar energy, wind power) but control is with human
 - However, in AI both power and control are built into AI itself.
 - So, AI can make decisions on its own – autonomous. It enables solutions like self-driving cars, UAV (+ve)
 - On the other, AI could supersede mankind. This is a technological singularity (-ve)
 - AI community does not talk of singularity, but man-machine coexistence.

Exercise 3

- Discuss the implications of the name “Artificial Intelligence”
 - This is the name given to the field of Intelligent Machines, which build natural intelligence into machines. The term Artificial Intelligence gives a sense of machine-generated or artificial form of intelligence. This is not correct. There Artificial Intelligence should be treated as a single term.
- Can you understand NaCl by considering the features of Na and Cl separately?
 - Although Na and Cl are poisonous, NaCl is a completely different thing, which is not toxic. (X, Y ☐ X, Y, X and Y, None)
- Will AI lead to reducing the intellectual capacity of mankind?
 - Unless AI itself supports the building of cognitive skills of humanity, there is a threat.

Early Developments in AI

- Since 1956, AI projects were funded by US Defense Ministry
 - Results were military secrets, unknown to public
- McCarthy (CMU) and Minsky (MIT) came to the Dartmouth conference in 1956. and McCarthy also joined MIT
 - Conflict McCarthy (intelligence is logic based), Minsky intelligence is based on anti-logic too
 - As a result, McCarthy went to Stanford.
- So, AI was developed separately at MIT and Stanford (with the supported of CMU)
- McCarthy (Expert Systems, Game, Self-driving jeep) - Minsky (NLP, Robotics)
- For more than 20 years development of AI was very slow
- “AI Winter” – No funding for AI and could not win the industry recognition

Early Developments in AI

- Early development was limited to logic/rule-based AI solutions (Because of McCarthy) and NLP by Minsky
- In 1984, Expert System, XCON won industry recognition for the first time after 30 years
- After 1990, CMU-based development in AI has been recorded (XCON, DART-Gulf War, Deep Blue, Pathfinder)
- Defense Ministry said 30 years of funding for AI recovered from DART
- All early developments come under symbolic AI
- Ingredients of these systems were *knowledge*, but not the *data*

Exercise 4

- Discuss, why AI was not popular until the late 1980s
 - Due to military intervention, AI findings were secrets and unknown to people
 - Since AI has power and control together people were afraid of AI
 - There was no industry recognition (XCON) until the late 1980s.
 - AI was not a science before the 1990s

Criticisms for AI

Artificial Intelligence is a misleading term

Natural Intelligence but not machine-generated intelligence

Can a machine think? (programs doing the task that requires thinking – game, problem-solving, theorem proving)

Can machine be creative? (arts, architectural programs, engineering)

Power and Control? (+ve, -ve)

Singularity (man-machine coexistence, Irobot,)

Military intervention (invisible to society for long)

Enthusiasm for AI

- XCON (1984) by DEC – used to configure mini-computers (gave recognition)
- DART (1991) – Expert system that executed the entire Gulf War
- Pathfinder (1997) – Rover explores the unknown environment in Mars
- IBM Deep Blue (1996) – Kasparov was defeated, and visualized few steps ahead, rule-based
- IBM Watson (2011) – Answering any question, rules + ML
- AlphaGo (2015) – more options than chess, so rules cannot be modeled, ML used
- Self-driving cars (2009) – Autonomy in AI, autonomous vehicles
- Tesla (2015) – self-driving cars/trucks
- Sophia (2016) – Citizenship in Saudi Arabia
- Amazon Alexa – work as a hub for controlling device at home
- Google Assistant – Answer any question
- Neuralink – Chip implanted in the brain
- ChatGPT (2022) - GenAI
- Boston Dynamics (1992-)

Homework

- Do an internet search on AI programs given in the previous slide, Hence identify their;
 - Origin/place
 - Application/Purpose
 - Contributors
 - AI technologies used
 - Intelligent features implemented

3. Diverse views on AI

62 views on life after death in Buddha's time

Atom is a concept with 3 diverse views

Dalton

Einstein

Schrodinger

- Neuroscience of Intelligence (two forms of intelligence)
- Analytical-based intelligence and Training-based intelligence
- Symbolic AI and Non-symbolic AI
- Artificial Cognitive Systems and Machine Learning
- Logic-based and antilogic based approach to intelligence
- Strong AI and Weak AI
- Generalized AI and Narrow AI
- Intelligence and Knowledge (CMU)
- Four-school of thoughts
- Turing Test and John Searle's Chinese room argument

What is an electron?

Wave

Particle

Both

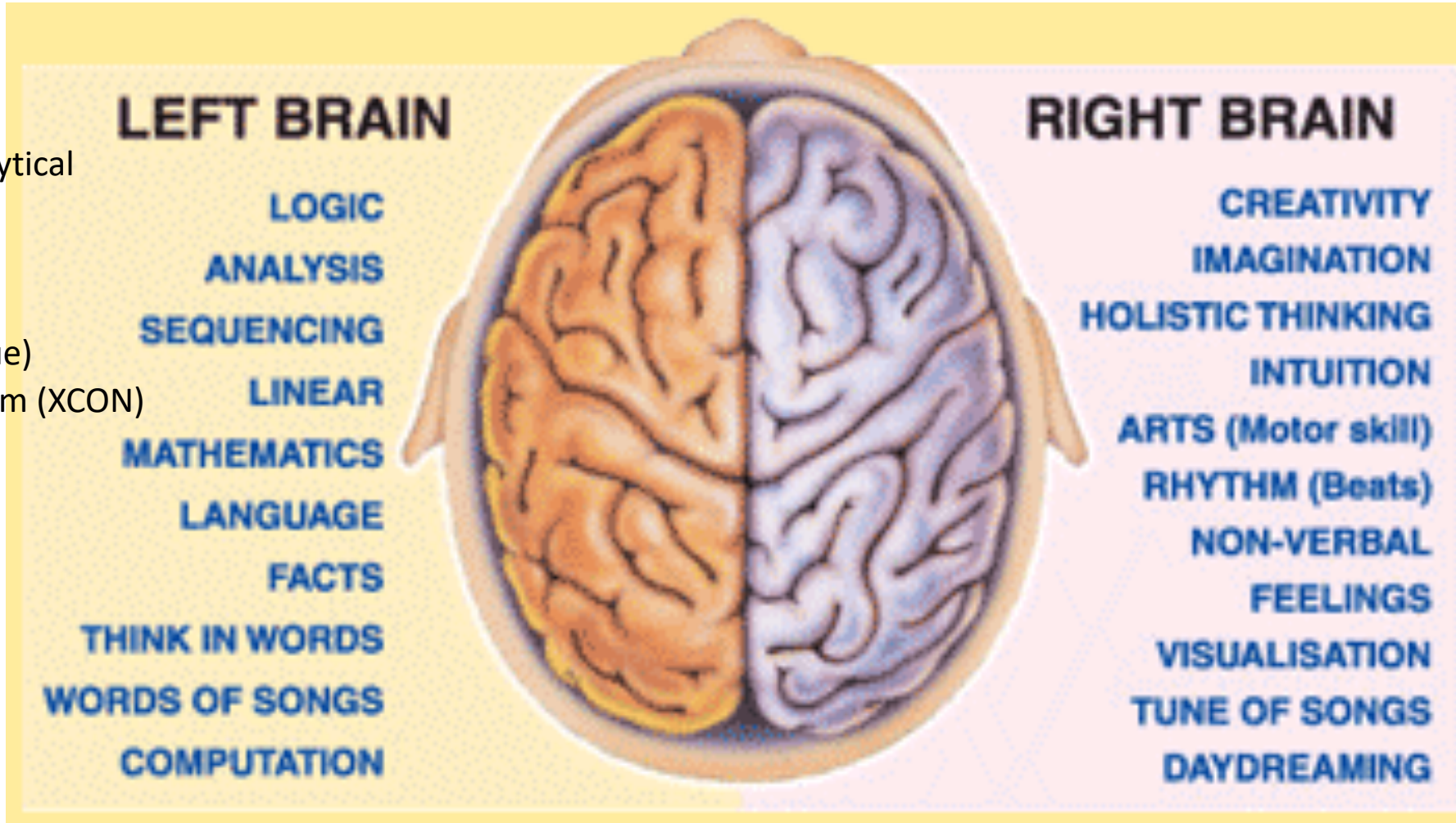
None

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Neuroscience of Intelligence

Thinking
Logical/analytical
Knowledge
Symbolic
Algorithmic
TT (deep blue)
Expert System (XCON)
NLP (Eliza)
Learning
Memory



Training
Training-based
Data
Non-symbolic (ML)
Non-algorithmic
TT (AlphaGo)
Expert Systems (Wa
NLP (chatbots)
Learning
memory

A Very lazy device?

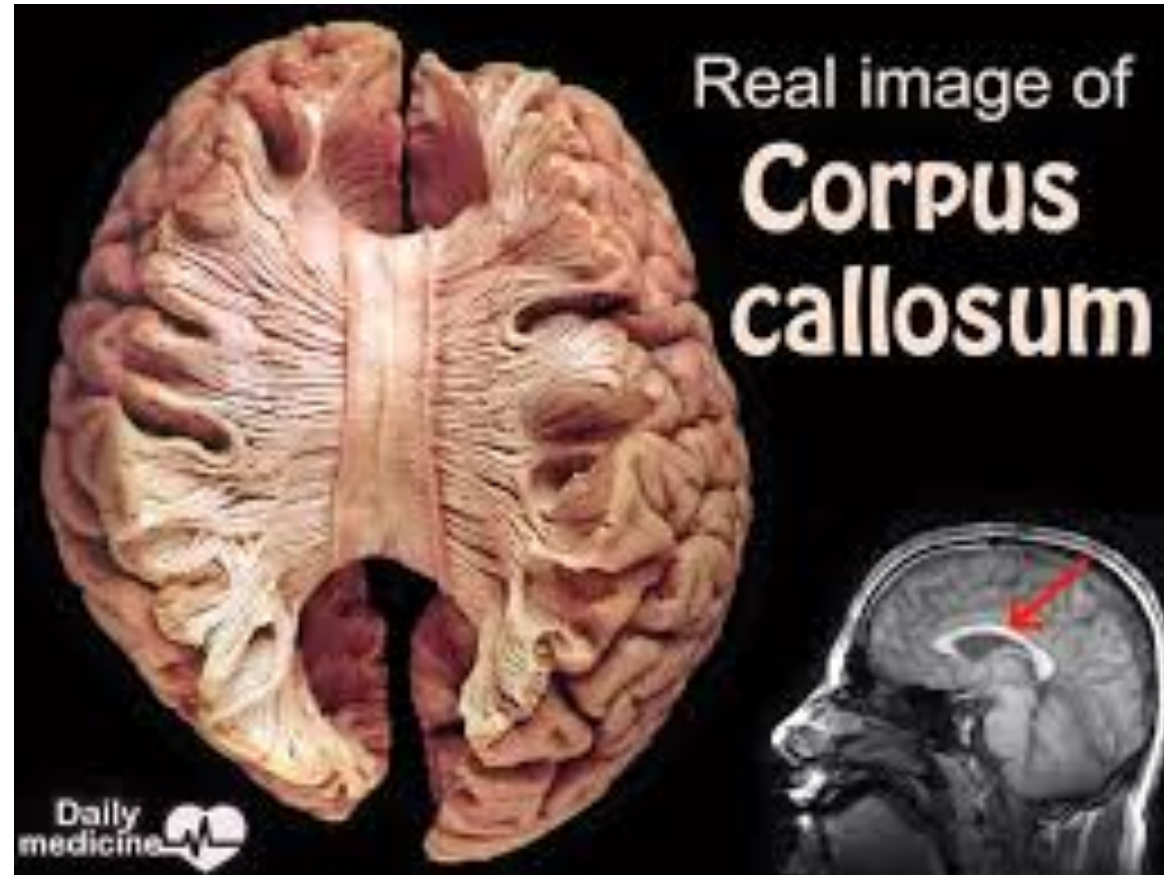
- The brain is a very lazy device
 - It has 100 billion neurons
 - But only 20 billion neurons are used (80% is unused)
 - Humanity got a thinking brain after millionaire evolution
 - But people do not like to think very much
 - Even using the left brain, we send some roles to the right brain
 - Learning Math also we training (multiplication tables/theorem...)
 - Walking is training-based and activates thinking only in unusual situations
 - Brain assigns takes as much as it can to the right brain
 - Brain tends to delegate tasks to peripherals organs (that is why hands, legs, eyes, etc. can move by training without thinking)
 - People still know very little about the brain

Analytical/Training based intelligence

- Analytical and Training are two major sources of intelligence in the brain
- This inspiration comes from Neuroscience
- Our intelligent activities are based on one or both of these forms
 - Human use both brains
 - Animal mainly use the right brain
- Preparation for an exam involves both forms of intelligence
 - Theory/logical/analytical study and training to solve past paper questions
- These two forms are mutually beneficial to each other
 - Theory can be better understood when we do practical (training)
 - Training for driving can be improved if we know what each paddle does
- Some professions are heavily based on one form of intelligence
 - Analytical – Mathematics, Engineering, Science,
 - Training - driving, carpenters, swimming, music,
- Animals are generally good at training-based intelligence
 - Dogs, elephants,
- When both sides are developed cross-connections are formed (Video)

Corpus callosum

- Evidence shows that mindful walking forms more connections in the corpus callosum
- When identifying left/right legs separately, it is an analytical task (left brain)
- When experiencing pain, hardness/softness in the foot (right brain)



Symbolic/Non-symbolic AI.....

- Symbolic AI refers to analytical/logical intelligence
 - Here intelligence/knowledge is symbolized by rules/logic, etc. ($S = ut + \frac{1}{2}at^2$)
 - Knowledge is the main ingredient for the left brain
 - Humanity becomes so intelligent after starting symbolic manipulation
 - Use of languages for communication is a part of symbolic manipulation
 - Computer is so powerful because we can communicate with computers through languages
 - Animals also become intelligent if they can communicate through languages (cows, dogs, elephants)
 - Symbolic manipulation is used in many subject areas
 - Mathematics, Science, Logic, Languages, Programming/computing
 - Symbolic AI is the oldest approach to building intelligent machines (why?)
 - DART, Deep Blue, pathfinder, etc. are symbolic AI-based intelligent solutions

Symbolic/Non-symbolic AI

- Non-symbolic AI refers to training-based intelligence
 - This area models the world using known data (examples)
 - Data is the ingredient for the right brain
 - The non-symbolic area is popularly known as machine learning
 - Suitable for areas with data but no theory/rules/algorithm, etc.
 - Many **recent developments** in AI are non-symbolic solutions
 - Watson, Google Assistant, ChatGPT, Atlas, etc. are some examples
- All AI techniques can be classified under
 - Symbolic
 - Non-symbolic
 - Both

Exercise

- Which of symbolic AI or non-symbolic AI is more important for understanding natural intelligence in machines?
 - Symbolic AI involves modeling intelligent features such as problem-solving, explanation, game, language manipulation, thinking,
 - The intelligence of humanity is based on symbolic manipulation.
 - Non-symbolic AI mainly models the intelligent feature called Learning (no explainability)

Artificial Cognitive Systems/Machine Learning

- Artificial Cognitive Systems (ACS) is known as the broad area of symbolic AI
- Machine Learning (ML) has covered a broad area of non-symbolic-AI
- Machine Learning was coined by Arthur Samuel (1950) and built a self-learning program to play checkers. Noted the more the system plays, the better the performance.
- Any AI technology can be classified under symbolic (ACS) and non-symbolic AI (ML)
 - ACS – Expert Systems, NLP, Gaming, Robotics,
 - ML – Machine Learning (ANN, CNN, RNN,
- Some AI techniques can use both symbolic and non-symbolic approaches to AI
 - Natural Language processing (knowing rules/grammar or training language can be manipulated)
 - Game playing (knowing rules or training game can be played)
 - Expert system (expertise could be due to knowledge of rules/theories or training)
 - Robotics (robots can work on rules/theories or training)
- Early developments in AI was dominated by ACS
 - XCON, DART, Deep Blue
- Recent developments were dominated by ML
 - Watson, Google Assistant, AlphaGo

Artificial Cognitive Systems/Machine Learning...

- Cognitive Science is the subject of all cognitive (mind) features including thinking, learning, communication, problem-solving, intelligence, perception
- But Artificial Cognitive System (symbolic-AI) we consider only the left-brain features
- As a result, there is confusion, because in AI we classify the cognitive feature of learning under machine learning (non-symbolic)
- But we know that cognitive feature learning can be done by not only the right brain but also by both sides of the brain
- Therefore, the current model of ML does not mimic the learning capacity of full-brain

Logic/Antilogic

- After establishing AI, McCarthy and Minsky started working at MIT
- McCarthy argued that intelligence is logic-based
- Minsky argued that intelligence is also antilogic-based
 - Mother settles a baby, knowing the stomach is full, adding salt into a curry, and deciding the first question to answer in the exam are not logical decisions
- McCarthy moved to Stanford and developed the area of Expert Systems
- Minsky from MIT introduced the area of Natural Language Processing
- All Stanford, CMU, IBM MIT contributed to the early development of AI with logic and antilogic (rules/frames) solutions.
- Note both logic and antilogic systems come under symbolic AI and involve knowledge

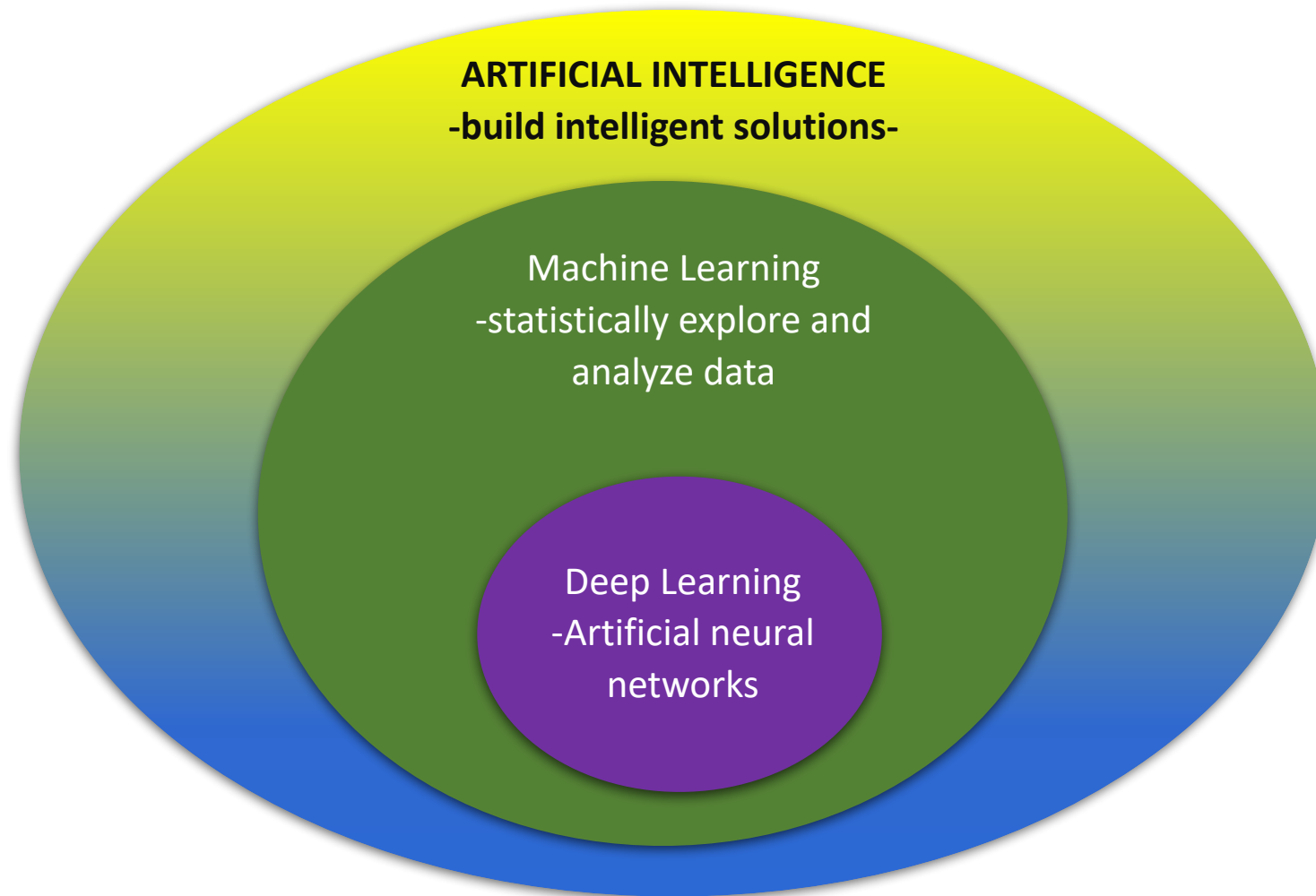
Exercise 5

- Why early development of AI was symbolic-based?
 - Early developments of AI were dominated by McCarthy's idea of intelligence (logic) and Minsky's idea of intelligence (antilogic), both come under symbolic AI. Both use knowledge as the ingredient of
- Why latest development of AI was non-symbolic (ML) based?
 - After the Internet (2000) so much data is generated via online transactions, social media, weather data, and hotel booking. Such data come in bulk noisy, incomplete, dynamic, real-time, non-algorithmic, unstructured, partial, etc. These data cannot be modeled by writing logical rules/algorithms. Further ML techniques (backpropagation, Deep learning) came up after 1990.

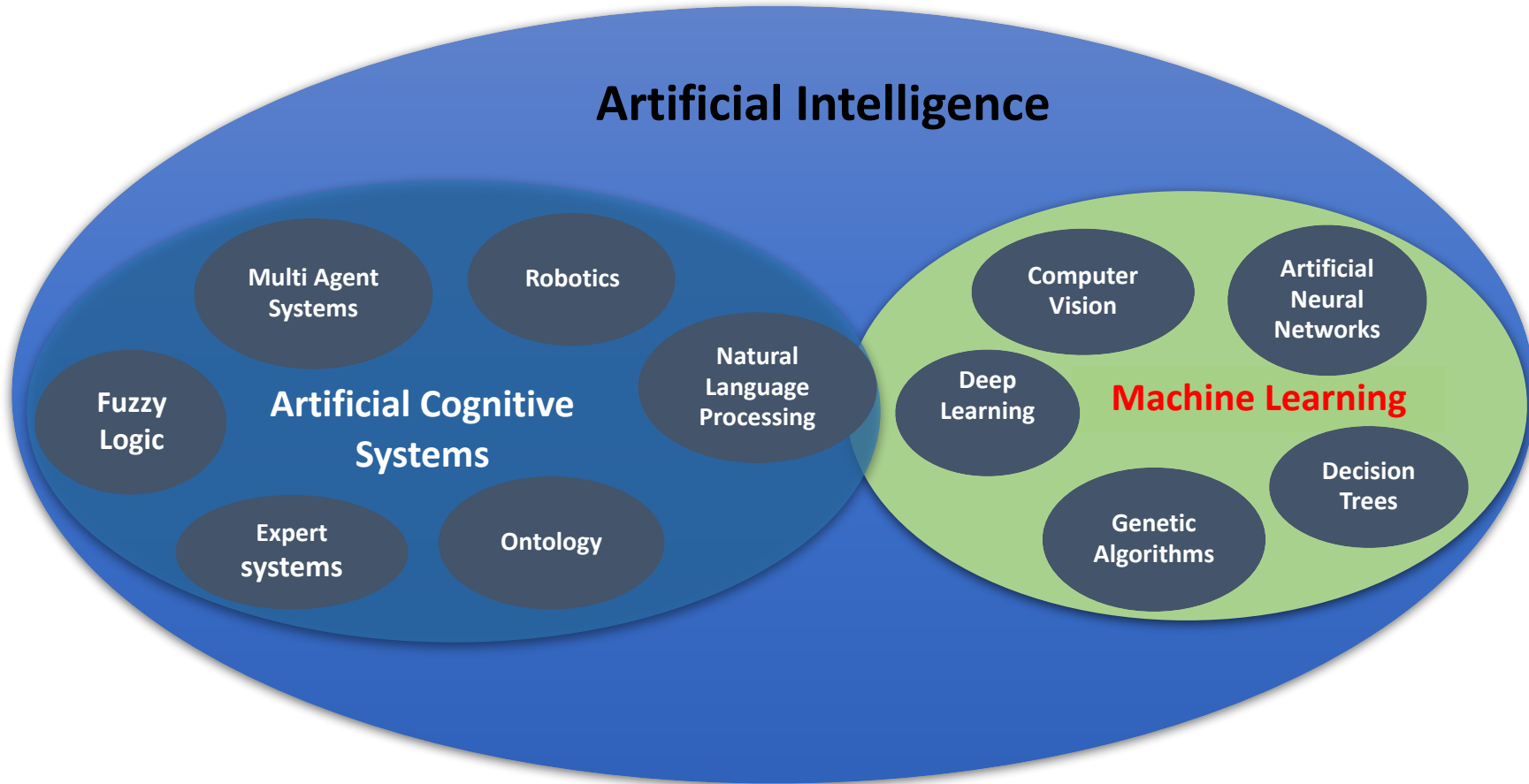
Exercise 6

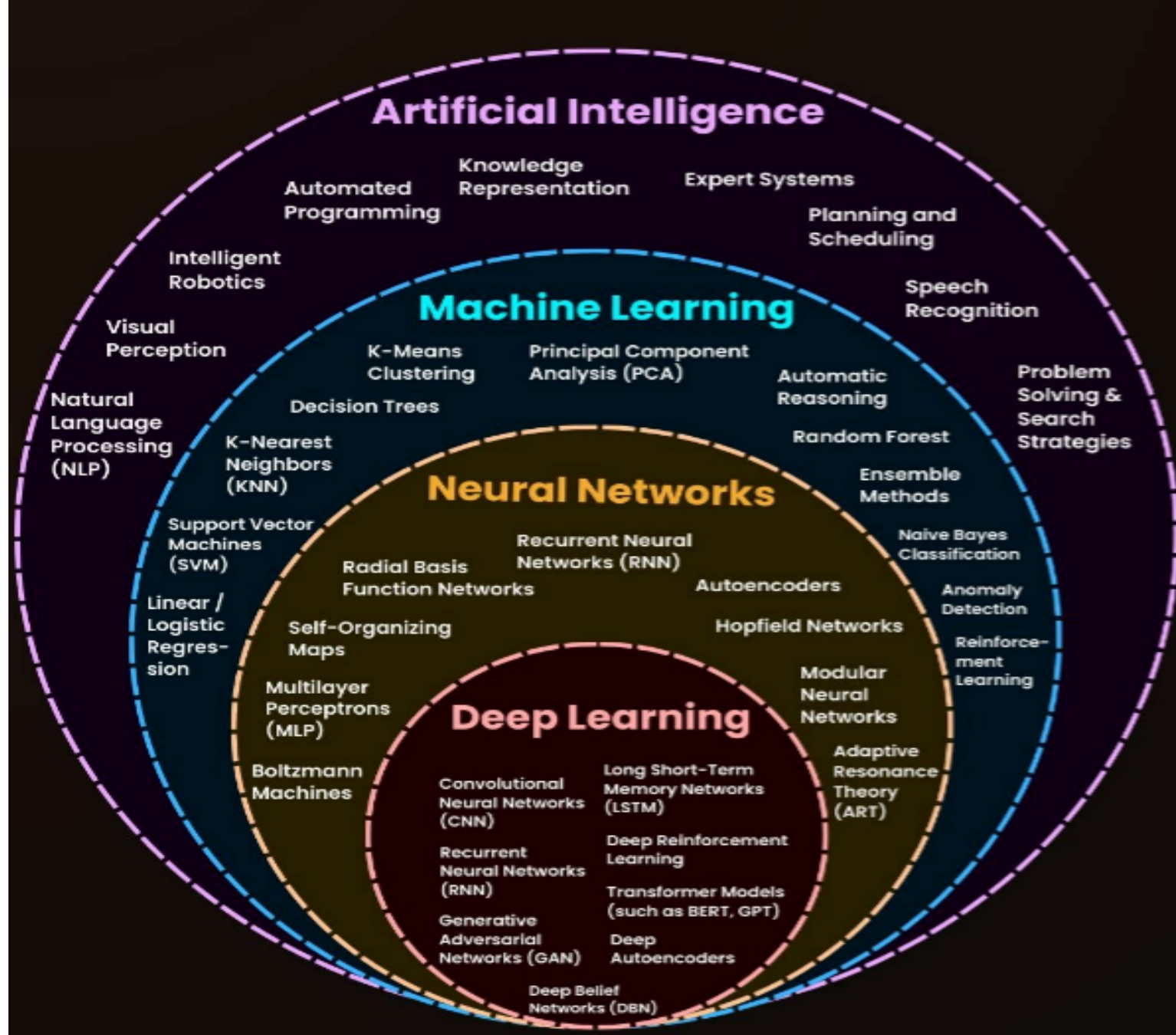
- Do you consider Machine Learning as a good model of how humans learn?
 - ML is about right brain, but learning is done by full-brain task.
- Can we consider rules a kind of logical knowledge all the time? No
 - All rule-based systems, procedures, laws, etc. constitutions are revised time to time due illogical things.
- Does logic surface the truth all the time? No
 - At the court of law, layers can locally argue and make even an innocent person is guilty and sent to jail.

AI and Machine Learning



Major Areas of AI





Exercise 6

- Discuss the implications of the terms
 - Artificial Intelligence
 - Artificial Neural Networks
 - Artificial Cognitive Systems
- It gives a sense of machine-created or artificial forms of intelligence, neural networks, and cognitive systems. But in all such cases, we have built models of natural intelligence, natural brain, natural cognitive features
- They were defined in the early days of AI. But later developments of AI have defined sub-areas of Intelligent machines as follows
 - Expert Systems, Natural language Processing, Multi-Agent Systems, Robotics, Machine learning, Machine Translation, Computer Vision, Ontology, Genetic Algorithms
 - The term artificial does not appearing in these

Exercise 7

- Why it is dangerous for machines to generate their own artificial form of intelligence? What are the steps taken to prevent this one?
 - A powerful nation might decide the kind of intelligence to be built into AI systems. Then humanity will not have the opportunity for further evaluation by many other communities in the world.
 - The areas like human-centered AI, Responsible AI, and Explainable AI are established to prevent the machine from creating its own intelligence beyond human intelligence.
- If a country does not allow people to use Facebook. What would happen to that nation?
 - People for alternate technologies. Therefore, that decision creates opportunities for developments in different directions.

Strong AI and Weak AI

- According to Weak AI, computers just simulate natural intelligence. This is what we normally do.
- In Strong AI, computers should implement real intelligence? This leads to the question of what is real.
 - What you feel is not real. Think of the example of going trip in a dream with all five senses working just like now.
 - So real cannot be defined
- What is wrong with weak AI?

Generalized AI and Narrow AI

Now three types of AI are discussed

- Artificial Narrow Intelligence (What we do now)
- **Artificial General Intelligence**
 - In the case of humans, we can give a problem without telling, which theory, model, or techniques should be used. But we can solve it. However, this is not the case with machines.
 - But early approaches to model intelligence: search and knowledge representation still work as generic approaches to solving any problem
- **Artificial Super Intelligence**

Intelligence and Knowledge

- AI started to model intelligence into machines
- But our understanding of intelligence is very limited
- However, we know that based on knowledge we can produce intelligence
- That is knowledge is the main ingredient of intelligence
- We gather knowledge from our early childhood
- So, people started to model knowledge to build intelligent machines
- This accelerated the early development of AI
 - Expert Systems, Natural Language Processing, Game Players, Robotics
- The last developments in AI are energized by Data.
- Future AI systems go beyond knowledge/data but use EEG signals directly from the brain
- Intelligence $\square$ Knowledge $\square$ data $\square$ EEG

Exercise

- Discuss how you capture the tree-climbing intelligence of a monkey to build a robot.
 - Intelligence and knowledge from monkeys cannot be captured
 - We can get some knowledge from animal scientists about monkeys
 - We can record the motion of monkeys as data when climbing trees. The approach used in Dynamics and Avatar

Four schools of thought

- There are four different definitions for the role of AI. It's a plus point
- Acting humanly (compared with a human)
 - Here, AI attempts to build machines that behave like humans. Many people understand AI in this manner. This is also what TT said. Expert systems (ACS) and Deep Blue (ACS), Google Assistant (ML)-like programs behave like humans. Here animal intelligence is disregarded.
- Thinking humanly
 - Here, AI attempts to build machines that can think like humans
 - Although thinking is a controversial term, we know tasks that require thinking: Problem-solving, Theorem proving, Game playing, etc. This is applicable for left-brain only

Four schools of thoughts

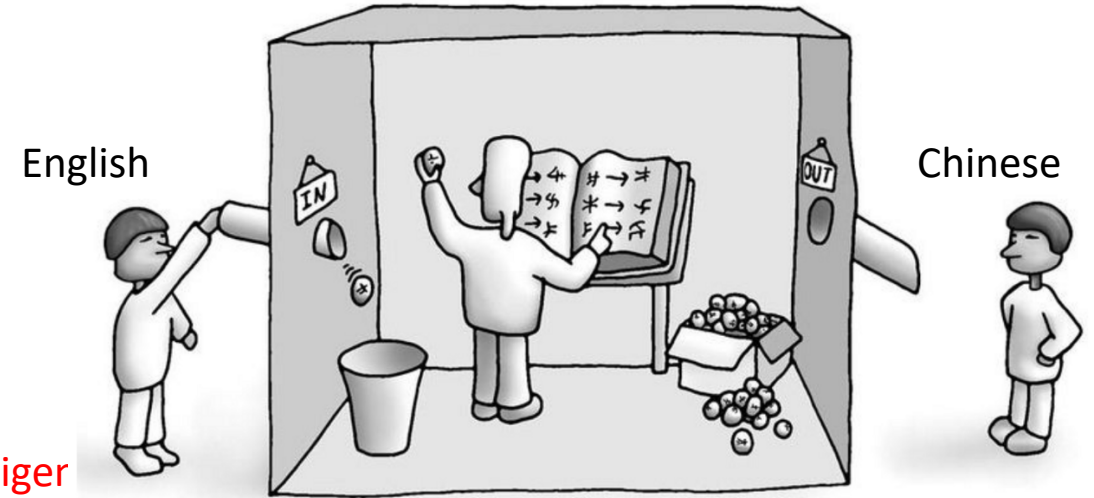
- Thinking Rationally (logically)
 - Here, AI attempts to build machines that can think logically
 - You cannot trust Logic all the time
 - $P \supset R, R \supset S \vdash P \supset S$: If the baby is dead, then the baby does not cry. If the baby does not cry then the mother is happy.
 - McCarthy said that intelligence is logic-based, we also developed many Expert Systems (PROSPECTOR) like programs using logic.
- Acting rationally (doing the right thing)
 - AI attempts to build machines that can do the right thing
 - Here we do not expect behaving humanly, thinking humanly, thinking logically but doing the right thing. – When a cobra enters a house, a rational servant attends baby first then the cobra
 - Modern definition/approach for AI known as building machines to do the right thing

Exercise

- Give an example, describe under what condition a robot that looks after kids at a preschool can be considered acting rationally (doing the right thing)
 - If a kid is not well, contact the doctor, inform the parent
- Discuss how Deep Blue could be considered under all four schools of thoughts
 - It works acting humanly because it can compete with a human (Kasparov)
 - It shows the ability to think like a human to select the most unexpected move
 - It shows thinking logically when following rules exactly
 - It acts rationally (doing the right thing) by purposely losing at some time.
- Under which school of thought the AlphaGo can be explained well
 - Comparing with a human (TT)
 - AlphaGo is trained using data which was generated by people when they play the game by thinking like human

Turing Test vs. John Searle's Argument

- According to the Turing Test, Intelligent Machines can be developed
- In contrast, John Searle argues that intelligence requires consciousness/thinking (being aware) . Machines just manipulate symbols without knowing what symbols mean. Therefore, computers cannot be intelligent
- He proposed the Chinese room argument to support this
 - The man in the room does not know Chinese or English
 - But he can take a note written in English and find the corresponding Chinese translation from the big book just by mapping symbols?
 - Do you say the man is intelligent?
- Although Searle argues that manner, people have already develop programs that require thinking.
- This is why, some argue that if a machine can behave as if it is intelliger why bother whether the machine does it consciously?
- When people do things without knowing what they are doing for a certain period, later they tend to be aware of what they do (exam, "golaya",)



Agent Technology

- Inspiration - Massive systems (solar system, brain, body, sun) are governed by computing taking place at the minute/tiny levels
 - Atoms (Sun $H \leftrightarrow He$)
 - Molecules ($2C_8H_{18}(l) + 25O_2(g) \rightarrow 16CO_2(g) + 18H_2O(g)$)
 - Cells ($C_6H_{12}O_6 + O_2$?)
 - Ants
 - Bees
 - Virus/bacteria/fungus
- This means any huge system can be modelled by tiny entities, we called them agents
- In computing we are interested in writing small programs as Agent
- This is closer to writing programs as classes/objects

Some characteristics of Agents

- Situatedness (works in a specific environment)
 - A dog is living at your home, it cannot work for another
 - A self-driving car is an agent that acts inroads
- Work for a master (Agent listens to a specific person)
 - A dog listens only to the master (if your dog listens to a neighbor the dog is not an agent)
- Autonomous (work with little/zero instruction, self-driving cars)
 - When somebody enters the premises dog barks without instruction from the master
 - Servant (agent) brings a cup of tea to a visitor
- Consume less resources (less facilities when agents run it uses less CPU/memory)
 - No special food, live outside the home

Some characteristics of Agents

- Create on request, execute and disappear (printer driver)
 - When visitor comes the dog wakeups, barks, get family members' attention then stop barking
 - Repairman came, attend the job, finish it, returns
- Adaptivity (adjustable to situation)
 - Once dog knows the visitor is a friend, dog stops barking
- Proactive (work without instructions)
 - From time-to-time dog walks around the garden
 - Servant cleaning, washing, watering,
- Reactive (suspend the current work and give priority to the master)
 - When dog chasing a person, but master talk the name of the dog, dog response
 - When the servant is on a call, master rings the bell, and then the servant attend the master.

Some characteristics of Agents

- Learning
 - dog become friendly with a visitor after some time.
- Able to move
 - Dog can walk within the house and garden
- Interact or sociability (cooperate, coordinate, negotiate)
 - When a dog barks at night, other dogs also bark
 - When a self-driving car operates on a road coordination among vehicle is required

Assignment

- Do a search on videos (10) on chronology/history of Artificial Intelligence
 - Compile a video to demonstrate the history of AI

4 Influential areas for development of AI

- AI is not a subject on its own but is influenced by developments in many other fields
- Philosophy (can a machine be intelligent)
- Mathematics and Statistics (formalizing, reasoning)
- Economics
- Neuroscience (understanding two major forms of intelligence)
- Psychology (mental behavior)
- Computer Science
- Computer Engineering
- Control Theory and Cybernetics (self-control and communication)
- Linguistics (NLP)
- Sciences (Physics, chemistry, biology, medicine,)

Philosophy

- Philosophy changes the direction of the world from time to time
- Dalton's atomic view changes in new directions – nuclear physics (Einstein), quantum theory (Schrodinger)
- Philosophy addresses many concerns such as life/death, real/simulation, intelligence/knowledge, brain/mind, man/machine, intelligent machines, computability, singularity, consciousness/awareness, autonomous, cost/benefits,
- Aristotle (343BC)– first to talk about how to draw conclusions/decisions
- Rene Descartes – mind/matter distinctions (dualism)
- Francis Bacon – philosophy of science (17th century)
- Newton - Scientific conclusions based on experimental evidence (Scientific Method)
- Philosophy is concerned with the fundamentals of intelligent machines.

Exercise

- Which philosophical concepts are relevant to AI
 - Real/simulation (Strong AI/Weak AI, John Seales), man/machine,
 - Intelligence/knowledge/data (we model intelligence using knowledge and data)
 - Singularity (can the machine supersede humanity ?)
- Is there any philosophical contribution from Turing to the area of intelligence machines?
 - Yes, The paper Computing Machinery and Intelligence
 - Computability with six parameters [?] AI becomes a programming task
 - Turing Test
- What are contributions by George Boole for mathematics/computing
 - Formalize Logic
 - Introduce Boolean algebra basis for digital electronics (1870 [?] 1948)

Mathematics and Statistics

- Mathematics can Formalize or Symbolize the world's knowledge
- Logic is the oldest way to formalize the knowledge
- Although Aristotle introduced Logic, George Boole formalized the logic
- Mathematics study on measurements, calculations, patterns
- Mathematics is a powerful system to do Reasoning/drawing conclusions
 - Science, Law, Engineering
- Reasoning under uncertainty(Statistics)
- Concept of Algorithm belongs to Mathematics, although it is used in Computing
- Gödel – limit of deduction (incompleteness theorem)
 - Like scientific theories mathematical theories can also be falsified.
- Allen Turing – Computability (which function can be computed)
- Tractability – solvable but grows exponentially with the size
 - Write all prime numbers between 0 and 10^{26}

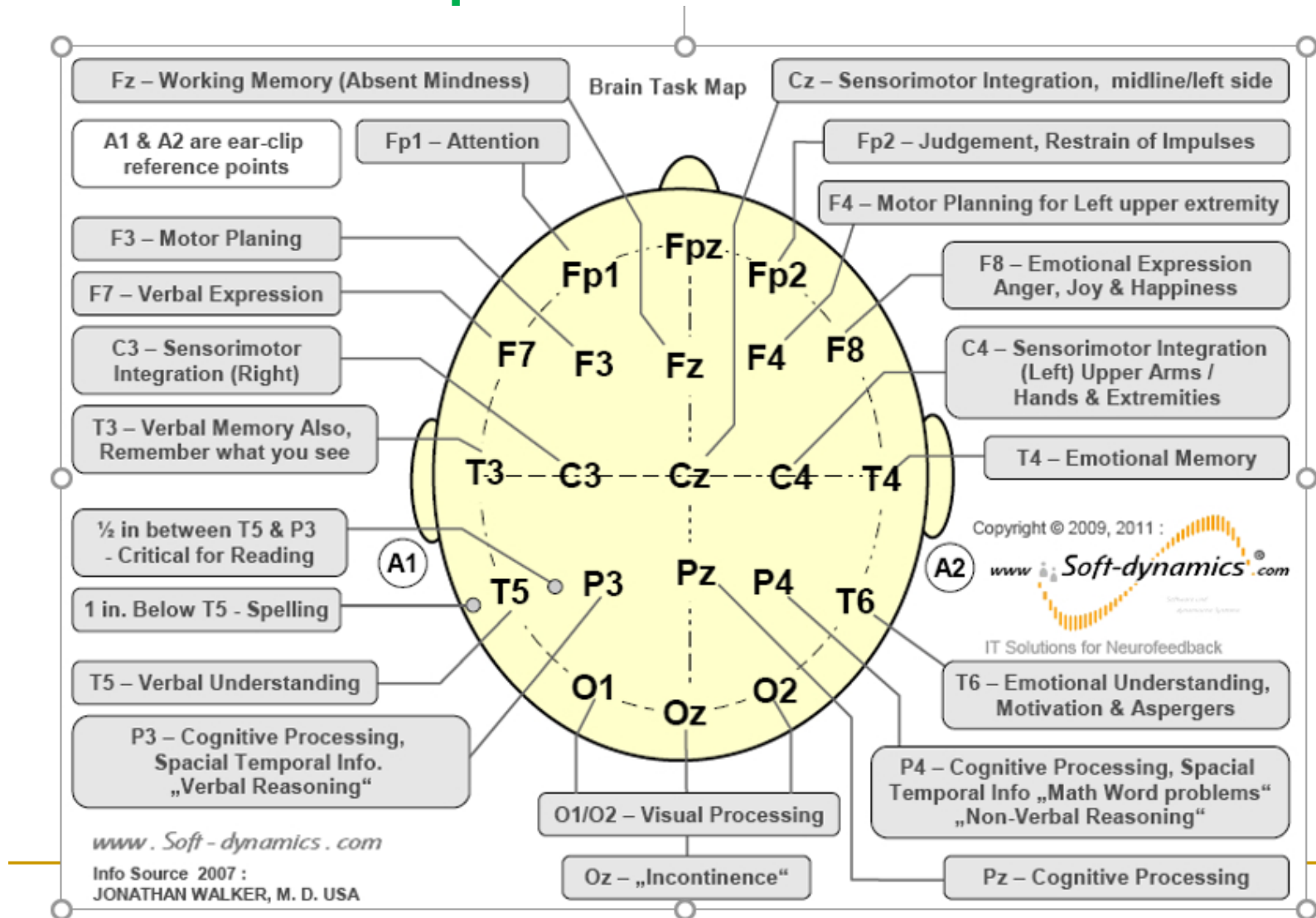
Economics

- Use of resources effectively
- it is not limited to money, land, etc. but even the choices of people
- Adam Smith (18th century) – first to treat Economics as a Science
- Concept of Utility
- Decision Theory (utility theory + probability)
- Game Theory
- Pioneer in AI, Herbert Simon won the Nobel Prize in Economics (1978)
 - Making good enough decisions is better than optimal ones with extensive calculations.
 - This idea closely goes with the modern approach to AI – building machines to do the right thing
 - Read Herbert Simon's book - The Science of The Artificial

Neuroscience

- Neuroscience is the Study of the nervous system, especially the brain
- Neuroscience gave the Insight for AI people to identify two forms of intelligence logical vs. training
- Foundation for symbolic AI and non-symbolic AI
- Neuroscience discovered
 - neurons and connections in the brain
 - Different areas in the brain are responsible for different tasks
 - when the brain performs different tasks, electrical signals (EEG) are emitted (Richards - 1870)
 - Techniques like EEG, fMRI, and CT have now enabled in-depth study of the brain
- Neuroscience will be the most influential area for the future of AI
 - Brain-machine interfacing
 - Biological programming (we can program organs, brains, etc.) - worms
 - Mind-uploading (EEG of an Expert [?] ANN [?] upload ANN to a brain) - puppy
 - Bionic/cyborgs (humans with extended physical/metal capacity – interfacing brain EEG with the devices)
 - Just like computers, biological systems will also be able to program.

Brain function map



Psychology

- The area is about how humans/animals think and take actions
- This area studies the human/animal behavior
- Cognitive Psychology, Cognitive Science, AI, and Behavioral science are related to psychology
- Cognitive psychology treats the brain as an information-processing device
- MIT pioneered establishing the field of Cognitive Science
- In 1966, MIT developed the first chatbot ELIZA to implement dialog between a psychiatric patient and a doctor
- ELIZA is one of the oldest AI programs under NLP

Computer Science

- Computer Science develops theory of computing, computability, tractability
- Computability can be demonstrated by paper and pencil without a computer
- This was done by Turing in 1936
- Theory of computing includes algorithms, data structures, programming languages, compilers, software engineering, Networking, ...
- Because of computer science, AI becomes a programming activity to build intelligent machines
- Since normal programming languages deal with numeric computations, AI community wanted special languages – Prolog, LISP, for symbolic manipulations (text, sentences, etc.)
- Nowadays, all Java, Python-like languages can also do symbolic manipulation
- Influence of CS for AI is noncomparable
- AI also contributed back to CS in parallel computing (ML), distributed computing (MAS), evolutionary computing (GA)

Computer Engineering

- To build efficient computers that can be programmed to be intelligent machines
- The Analytic Engine by Babbage (1832) was the first model of a computer (high speed)
- Analytical engine is a mechanical computer
- Ada Lovelace wrote the first program for the incomplete Analytical Engine of Babbage
- The idea of a machine for computation was proposed by Turing (1936)
- In 1940, the first operational computer was developed by Turing's team to decipher German messages
- This computer was made of vacuum tubes
- After the invention of the transistor (1948) the modern digital computer was invented
- Modern computer architecture was proposed by Von-Neumann
- Hardware technologies like Memory, CPU, Networking, GPU, and Cloud have developed much faster than software technologies
- However, AI as a set of software technologies has changed the entire computing landscape.

Control Theory and Cybernetics

- How artifacts can work under their own control - power + control (self-driving, autonomous, making their own decisions) has been a concern for people for many centuries
- The first self-controlling machine was the water clock by Ktesibios (250BC)
- Thermostat (17th Century) and Steam engine governor (18th century), Pythagoras Greedy Cup (570BC) and are also self-control machines
- Today, AI has developed autonomous artifacts by programming, while all the above are mechanically powered.
- Wiener defined cybernetics as “the science of control and communications (Internet) in the animal and machine.”
- In the future, BCI would enable direct communication/control between man and machine

Linguistics

- Linguistics is about the relationship between language and thoughts
- Computers become so significant because we can communicate with computers by using programming languages initially, and nowadays using natural languages like English,
- Modern linguistics and AI grew up together as Computational Linguistics or Natural Language Processing
- Apart from autonomous solutions, NLP (LLM) has revolutionized the developments in AI in the recent past (ChatGTP).
- The behavioral approach to language learning is not accepted by Norman Chomsky
- Behavioral theory cannot explain notion of the creativity - it did not explain how a child could understand and make up sentences that he or she had never heard before.
- The problem of understanding language has still become more complex than it seemed in 1957

Natural Sciences

- Scientific method
 - Crust of SM is the use of experiments to generate data and to prove things
 - Statistics is used to express the significance of conclusions
- Physics (Michio Kaku)
 - Digital Computer
 - Quantum Computer
- Biology
 - Genetics (GA)
 - Neurosciences (ANN, ML)
 - Cyborgs (humans with technologically extended physical/mental capacity, Bionics)
 - Robotics (human/animal behavior – Boston Dynamics)
- Chemistry
 - Complexity Science (MAS)

5. History (Developments in) AI

- Knowledge of the development of a subject, one can build enthusiasm for the subject
- When you are in the present, knowing the past you can imagine the future (research)
- Listen to Yuval Harari about AI and history
- Below are the milestones of the development of AI
 - Gestation of AI (1943-1955)
 - Birth of AI Early (1956)
 - Early development (1952-1969)
 - Dose of reality (1966-1973)
 - Knowledge modeling (1969-1979)
 - AI becomes an industry (1980-present)
 - Return of Neural Networks (1986-present)
 - AI adopts Scientific Method (1987-present)
 - The emergence of Intelligent Agents (1995-present)
 - Availability of very large data sets (2001 – present)
 - The State of the art

Note on Assignment for Intelligent Machines

- Mechanical
- Electrical
- Electronics
- System
- Process

- Blub ☐ Intelligent Bulb
 - Identify where human intervention is required
 - Switch on when dark
 - Switch off when light
 - Adjust intensive according to the level of light
 - Switch off when no motion/unattended for some time

History (developments) of AI

- **The Gestation of AI (1943-1955)**

- In 1936 Turing Machine was introduced to show computability
- After World War 2 lots of enthusiasm to discover new knowledge
- Curiosity about intelligent machines
- In 1940, Turing's team deciphered German messages by developing a computer
- In 1943, McCulloch & Pitts - mathematical model of the human brain (ANN)
- In 1948, Donald Hebb developed the first algorithm for ANN training.
- Von-Neumann's (1945) architecture for computers was introduced
- Turing's paper (1950) on Computing Machinery and Intelligence
- Note: The Turing Test was also included in that Paper
- In 1948 transistor was invented, and the digital computer was born
- In the 1950s, John McCarthy and Minsky completed their Ph.D. in the Department of Mathematics at Princeton University
- After PhD, McCarthy Joined Carnegie Mellon (CMU), and Minsky joined MIT
- Note: Einstein (nuclear physics) and Turing (computer called Bombe) were very influential during World War 11.

History (developments) of AI

- **The birth of AI (1956)**

- Dartmouth conference (10 attendees) – proposal for the distinct subject area for intelligent machines by McCarthy (CMU).
- McCarthy coined the name Artificial Intelligence for the field of intelligent machines
- McCarthy defined AI as the Science and Engineering of building intelligent machines, especially programs
- This means AI strives to understand natural intelligence and build intelligent machines
- McCarthy joined MIT (where Minsky was) then Stanford
 - McCarthy - Intelligent is logic-based
 - Minsky - intelligent is anti-logic based as well
- Most of AI initiatives were funded US-defense ministry under DARPA
 - Findings were used for weapons, and findings were military secrets
 - No wars to test the power of AI-based weapons
 - As a result, AI was not known to ordinary people for 30 years

Developments in AI

- **Early enthusiasm (1952-1969)**

- After establishing the field of AI, the challenge was to develop intelligent machines/programs
- But for this purpose, people have very little understanding of natural intelligence
- Due to McCarthy many people believed that intelligence is logic-based. As a result, many logic-based programs were developed in the early days
 - General Problem solver (CMU)
 - Game playing (CMU/IBM)
 - Theorem provers (IBM)
- In the Early 1960s, Minsky Criticized the ANN model (1943) saying it was useless
- People gave up research in ANN for nearly 25 years
- Search algorithms are identified as the 1st approach to model intelligence
- Because people realize the fact that searching requires intelligence
- During this time (1952) Arthur Samuel demonstrated the concept Machine Learning

Developments in AI

- **Knowledge Based systems (1969-1979)**

- People realize knowledge as main ingredient of intelligence
- Knowledge is everywhere: learning, books, papers, web,
- Modelling knowledge instead of intelligence (Allen Newell, CMU)
- Knowledge modeling/representation is the 2nd approach to AI
- Many knowledge-intensive AI programs were developed
- Natural Language Processing (ELIZA – first chatbot by MIT 1964)
 - Dialog between a psychiatric patient and a doctor
- Natural Language understanding - SHRDLU (MIT 1968)
- Birth of Expert systems: DENDRAL, MYCIN (Stanford, CMU)
- Minsky at MIT (1975) proposed Frames to model knowledge

Intelligence [?] knowledge [?] data [?] Brain waves (EEG)

Developments in AI

- **AI becomes an industry (1980-present)**
 - XCON (1987) Expert Systems to configure mini-computers for DEC (CMU)
 - Fifth generation project by Japanese (failure)
 - Return of ANN (1986) with Backpropagation algorithm
 - Turning point in the new area of Machine Learning
 - AI Winter 1988 – No further funding for AI (especially DARPA was upset)

Developments in AI

Author Samuel

Minsky's criticism (1960)

Backpropagation (Rumelheart and MaCland) 1986

Deep Learning (Geoffrey Hinton) 2001

- **AI becomes a science (1990s-)**
 - AI was not a science for 35 years – theories were not supported by experiments
 - Science has been regarded as the most powerful body of knowledge
 - Scientific theories are developed based on Experiments
 - Expandability is a distinct power of Science
 - After the 1990s, AI started to use the scientific method to develop theories
 - Making hypothesis
 - Setting experiments and generating data
 - Statistical analysis of data from experiments
 - Data [?] Theory $F=ma$ (knowledge)
 - The process from hypothesis to Theory is called research
 - In science we draw conclusions based on a large number of experimental data
 - When AI becomes a Science, people start to trust AI

Developments in AI

- **Emergence of Intelligent Agent (1995-)**
 - A new approach to design and build AI software
 - Coding the software as Agents (like objects)
 - Agents are small codes doing specific tasks
 - They are like bees, ant,
 - Allen Newell, Laird, and Rosenbloom, proposed the first known complete Agent architecture (CMU, 1987)
 - A new approach to realizing Artificial Intelligence strives to build intelligent Agents.

Developments in AI

- **Availability of very large database (2001-present)**

- Computer Science emphasized on algorithms for more than 60 years
- But nowadays, data has become more influential than algorithms to make decisions
- Because of the Internet, so much data is generated by online transactions, social media, and online business,
- These data include words, images, photographs, voice, and video, which are rather noisy, incomplete, partial, non-algorithmic, etc.
- Machine learning was reborn with a new sharp (Deep Learning) to model such data
- Crafting on knowledge sources to build algorithms becomes less important than machine learning methods

10

27/08/2024

Exercise

- Discuss reasons for not recognizing AI for about 30 years. Also, describe how AI gained popularity after that.

From the beginning, AI projects were funded by the US Defense Ministry. They are projects on weapons, finding were secrets, that could not be tested as no war after World War 11, so people were unaware of the development of AI. Some people are afraid of AI due to power and control issues. AI was not a science, no industry demand for AI. Due to a lack of findings/results, AI faced AI winter.

XCON Expert system won industry recognition (1988). Rebirth of ANN, AI-based weapons were used in the Gulf War, the entire Gulf War was done by an expert system called DART, Deep Blue defeated Kasparov,

The state of the Art

- **Autonomous control:**
 - Autonomous Land Vehicle in Neural Networks (ALVINN) (1989) computer vision system (by NAVLAB of CMU) to steer the car to keep it following a lane
- **Logistic Planning Expert System:**
 - Dynamic Analysis and Replanning Tool (DART) (1991) by DARPA for logistic planning during the Gulf War
 - DART could handle more than 50,000 entities of tri-forces, weapons, vehicles, civilians, ambulance, medicine, water,
 - US defense ministry said that they recovered all investments for AI research for 30 years through the DART program
- **Autonomous planning and scheduling**
 - NASA has developed several remote agent programs for on-board autonomous planning (Jonsson, et al 2000)
 - Mars Pathfinder (1996)

The State of the Art....

- **Game playing**
 - IBM Deep Blue (1997) defeated the chess grandmaster. IBM's stock increased by \$18 billion.
 - Started at CMU in 1985 under the name Chip Test
 - Deep Blue could visualize multiple steps ahead beyond human capacity
 - Note Deep Blue is a rule/logic-based symbolic-AI solution
- **Robotic Vehicles** (driverless vehicles)
 - First Driverless robotic car STANLEY by CMU (2005)
 - Tesla, Google self-driving
- **Future direction?**
 - AI should be developed to improve machine intelligence beyond human intelligence
 - AI should be developed to improve human intelligence

The state of the Art...

- **Robotics:** iRobot corporation sold over two million Roomba robotic vacuum cleaner. Humanoid robots are also used in the kindergartens to elderly homes. Medical surgeries are now with robots. Further robots attend hazardous, explosive, etc.
- **Machine translation:** Natural Language Processing, understanding, and translation have recolonized modern AI. The initial symbolic AI approach to those areas has now become a machine-learning exercise. Chatbots are popular for interacting with people
- **Latest developments:** AlphaGo, Watson, and Sophia are ML based
- The concept of the Transformer has made it possible to read documents beyond linear reading. Therefore, documents with thousands of pages can be searched and analyzed very quickly. This is how ChatGPT became a champion of handling large language models (LLM). Key latest development in AI comes under NLP.

6. Major AI Technologies

- Expert Systems
- Artificial Neural Networks
- Genetic Algorithms
- Fuzzy Logic
- Multi Agent Systems
- Robotics
- Ontology
- Natural Language Processing
- Machine Learning
-

Expert Systems

- Expert Systems (ES) is a model of how human experts solve problems
- It separates Domain knowledge from Problem-solving knowledge
- Champion of symbolic AI
- ES is the first AI technology to give industry recognition (XCON)
- Features of Expert Systems
 - Explanations for solutions
 - Alternative solutions
 - Handling incomplete information (ask questions, assume, ignore)
 - Handling Uncertainty
- ES gives recommendations rather than exact answers
- Suitable for diagnosis, advisory, scheduling, planning, and design problems
- Not suitable for time-critical applications, why?

Exercise

- Which of the following problems can be modeled as an ES? Give reasons
 - Library catalog system -NO
 - Student's' information system for a university
 - Travel advisory services
 - Online shopping
 - Flight reservation -
 - Arithmetic Calculation – $8+7$
 - Missile guidance system

Artificial Neural Networks

- Model of the human brain as a network of neurons and connections
- Model of Learning from data?
- ANN is the ancestor of Machine Learning and Deep Learning
- It is the champion of non-symbolic AI
- ALVINN is the first ML solution to automate self-driving Jeep
- Longer time for training but less time for answering
- ANN can learn from non-algorithmic, incomplete, noisy, dynamic data
 - Weather data, hand-written documents, facial recognition, online billing, traffic data,
- ANN is ideal for the following tasks that can be done by human
 - classification
 - Recognition
 - Predictions
 - construction
- ANN gives answers, but cannot provide Explanations

Exercise

- Which of the following problems can be modeled by ANN
 - Signature recognition for Bank - yes (just data, no algorithms)
 - Loan advisory system – No (need explanation)
 - Traffic control – Yes
 - Flight reservation – Yes
 - Missile guidance system -
 - Sorting fruits into baskets on a farm
 - Weather forecasting
 - Disaster management

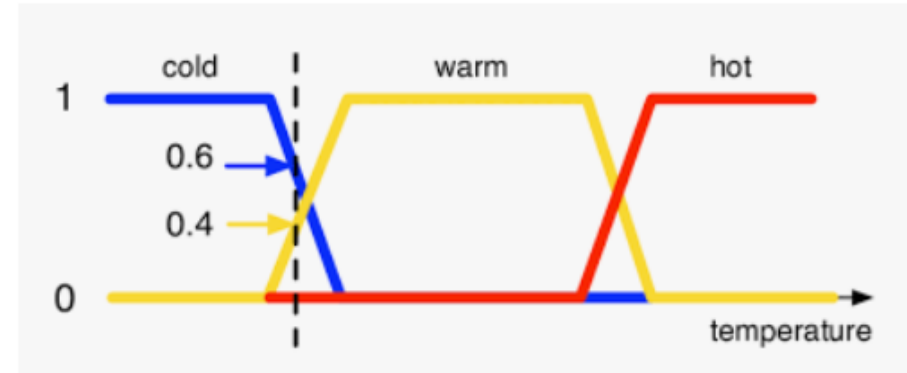
Genetic Algorithms

R	D	T
B	F	M
B	P	T
S	D	M

- GA is also a machine-learning technique
- It's a model of chromosomes and genes in biological systems
- Chromosomes undergo genetic operations: cloning, cross-over, mutation
- Cross-over produces children-chromosomes with similar features to parents
- Mutation produces children-chromosome with different features to parents
- GA ensures genetic operations on high-fit chromosomes (survival of the fittest)
 - Best quality solution (evolve) are generated over the generation
- GA can model real-world problems with dynamic data (better than ANN)
 - scheduling/planning, design, optimization problems
- GA can model with less data or no data (ANN requires lots of data)
 - Randomly/trial and error, assign some data and check they meet the fitness condition
- GA has given birth to the area of Evolutionary Computing

Fuzzy Logic

- Some real-world problems can accept many answers within a range
 - stopping a car at the traffic light
 - adding salt to a curry
 - controlling device (drones, fan, camaras,..)
 - Weight of cloths for WM
- Fuzzy logic (FL) systems are also modelled as symbolic-AI solutions
- Fuzzy logic workout a membership function to model a solution
- FL can also explain solutions like expert systems
- FL can also work with ES, ANN, and NLP to offer hybrid AI solutions



Multi-Agent Systems

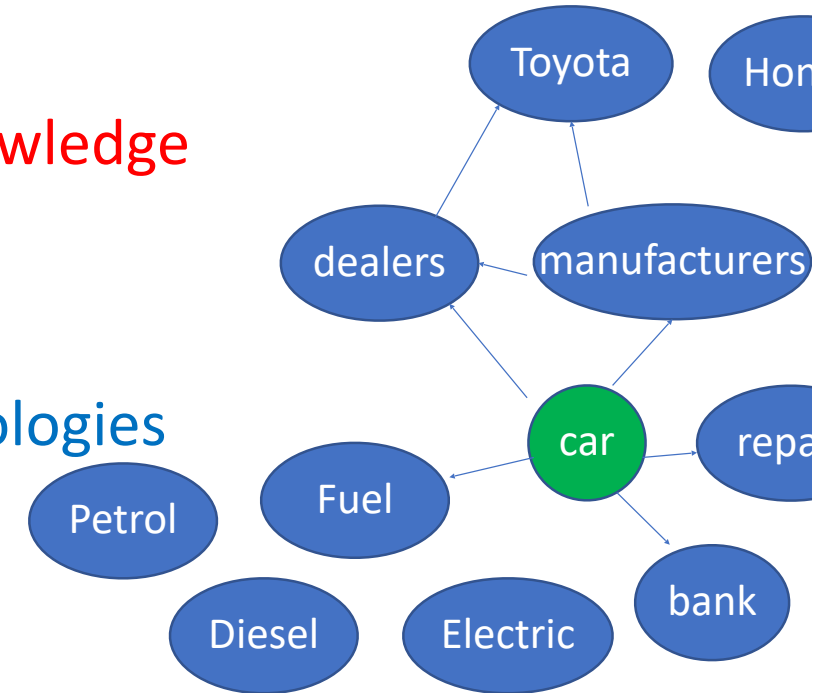
- Model of problem-solving by teamwork (discussion, message passing)
- Message passing is more effective than algorithmic problem-solving
- This involves communication, coordination, negotiations
- Inspiration from the behavior of ants, bees, bats, birds, etc.
- Mainly implemented as symbolic AI solutions
- MAS can solve problems with less (even without) data/knowledge
 - Initial stage of COVID
 - Disaster management
 -
- MAS can simulate any engineering, social science, biological, physical system
- NetLogo

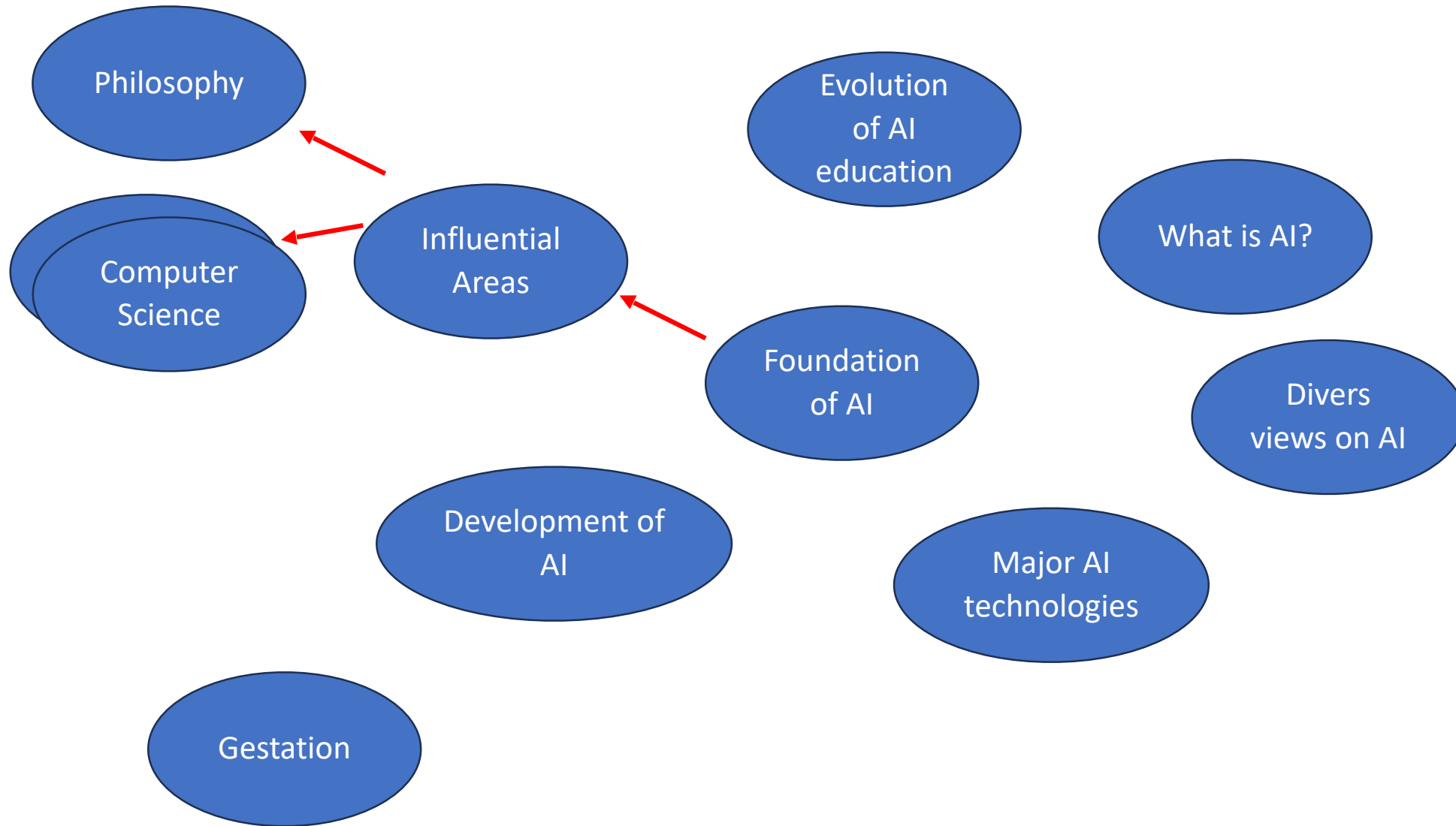
Robotics

- Implementing human/animal-like machines (e.g. Boston Dynamics)
- Robots have sensors to perceive the environment, and actuators/effectors return the output.
- Robot can use ES, ANN, GA, MAS like technologies to solve problems
- They can also use knowledge or data to solve problems
- Robots can work in environments where people cannot reach
- Robots are generally autonomous to a greater extent
- Robots range from nanorobots, households, industry to huge spacecraft
- There are the types of robots
 - Manipulators (grippers, assembling, manufacturing, they do not move)
 - Mobile Robots (navigate – self-driving car, autonomous vacuum cleaner, UAV,)
 - Humanoid robots (from kindergarten to elderly people)

Ontology

- Ontology is a powerful way to organize any form of knowledge
- Our knowledge is naturally organized into an Ontology
- Ontology comprises entities and relationships
- We can communicate only when we have sharable ontologies
 - Doctor and Farmer ???
 - Doctor and Nurse
- Learning is also a process of building/editing ontology
- Developed ontology are available for domains like medicine, vehicles, plants, animal, etc.
- Ontology can be used to model unstructured/incomplete knowledge





Natural Language Processing

Probably the oldest AI technology after the birth of AI (1st chatbot ELIZA). NLP model language processing, understanding, translations, construction' Initially NLP was developed under symbolic AI. Now NLP can also model by non-symbolic (machine learning). Because we can handle languages by knowing grammar or/and also by training' NLP is now in AI systems such as Expert Systems, Chatbots, etc. has revolutionized the modern developments in AI. ChatGPT (large language models).

Transformer is the concept that handles LLM efficiently

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- NLP model language processing, understanding, translations, construction,
- Initially NLP was developed under symbolic AI
- Now NLP can also model by non-symbolic (machine learning)
- Because we can handle languages by knowing grammar or/and also by training
- NLP is now in AI systems such as Expert Systems, Chatbots, etc.
- NLP has revolutionized the modern developments in AI
 - ChatGPT (large language models – LLM)
 - Transformer is the concept that handles LLM efficiently
 - Transformer allows to analyze the text beyond sequentially reading, fast

Machine Learning

- ML is one of the broad areas of Artificial Intelligence
- First ML model, ANN, was developed even before the birth of AI
- The ability to learn from non-algorithmic data is the power of ML
- Generally, a large amount of data is required to develop an ML solution

Blood pressure	Blood sugar	Lipid profile	Exercise	Sleeping	Weight	Hight	Stress	Health
90-120	100	high	yes	8	80	2	low	good
70-100	130	high	no	6	70	1.5	high	Bad
70-120	90	low	yes	16	80	1.8	high	Good
80-140	Missing	low	no	8	60	2	low	Bad
60-100	80	high	yes	7	80	2.1	low	???

Machine Learning

- The same data set can be modeled by more than one ML technique (ANN, GA, DT (decision trees))
- These techniques encode the data into different structures
 - ANN – Network, GA-one dimensional array, DT - Tree
- Machine Learning (ANN) has many areas like Deep Learning, Artificial Neural Networks, Convolution Neural Networks, Recurrent Neural Networks, Reinforcement Learning
- The choice of ML algorithm depends on the nature of the data
 - Artificial Neural Networks – Numeric Data
 - Convolutional Neural Networks – Graphics/images data
 - Recurrent Neural Networks – Sequential data (text/audio/video data)
 - Reinforcement Learning – Learning from the environment (game/self-driving cars/packaging)

Demand for AI solutions

- AI solutions are specifically applicable where humans have limitations
- Requiring to handle so many parameters at once in decision-making
 - Weather, agriculture, wars (weapons, soldiers, location, food, medicine, water, army, navy, Airforce, civilians, casualties,)
- When dealing with unknown environments/area
 - Online shopping, seeking medical/legal advice
- When humans are too biased/busy/unfit to make decisions
 - Unmanned vehicles, application processing
- When requiring a small reaction time for decision-making
 - Real-time systems, medical surgery
- When involving biases, accuracy, excitement, fatigue, etc.
 - Disasters, emotions

Concerns for building AI solutions

- Nature of domain
 - Algorithmic (symbolic)/Non-algorithmic (non-symbolic)
- Availability of Rules/theories or Data
- Answer only vs. Requirement of Explanations
- Reaction time vs. Concerns of Options
 - Speed, Memory, Reaction time
- Both ACS and ML can model some problems
- Urgency – Development time
- Maintenance and updates